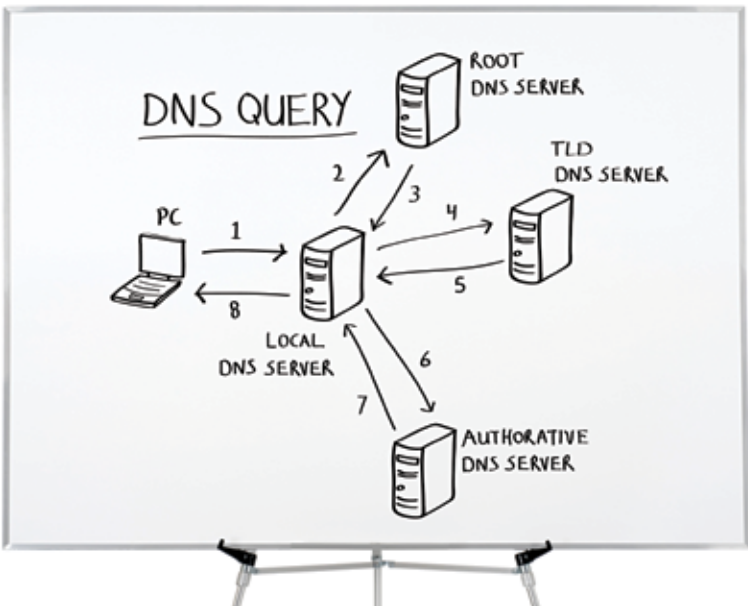


be communicated (mapping, routing, etc). The various protocols that we are about to go over briefly have one overarching theme - they specify the standard for how communication is packaged. The transmission is left to the higher abstraction levels of the TCP/IP.

DNS

The Domain Name System is a protocol for mapping resources between the two major namespaces - domain names and IP addresses. IP addresses are the equivalent of physical locations of websites (technically, of the servers that host the websites) on the internet, while domain names are easily recognizable labels that refer to those locations. This reference is what is taken care of automatically and efficiently by DNS. It essentially provides a worldwide, distributed, decentralized and scalable directory system that makes it possible for websites to have human-readable URLs. When you enter a url into your browser, the DNS Name Server directs the request to the appropriate IP address, according to the domain name entered by the user. However, the DNS Name Server also resides on a computer connected to the internet, so the IP address of the name server



DNS is a hierarchical system.